

1. Object Detection Performance Comparison

Given Data

Model	True Positives (TP)	False Positives (FP)	False Negatives (FN)
Model A – Viola-Jones	80	20	30
Model B – YOLO	90	15	20

(a) Calculate Precision and Recall

Model A – Viola-Jones

Precision:

Recall:

Model B – YOLO

Precision:

Recall:

Results

Model	Precision	Recall
Viola-Jones	80%	72.73%
YOLO	85.71%	81.82%

(b) Comparison

YOLO has:

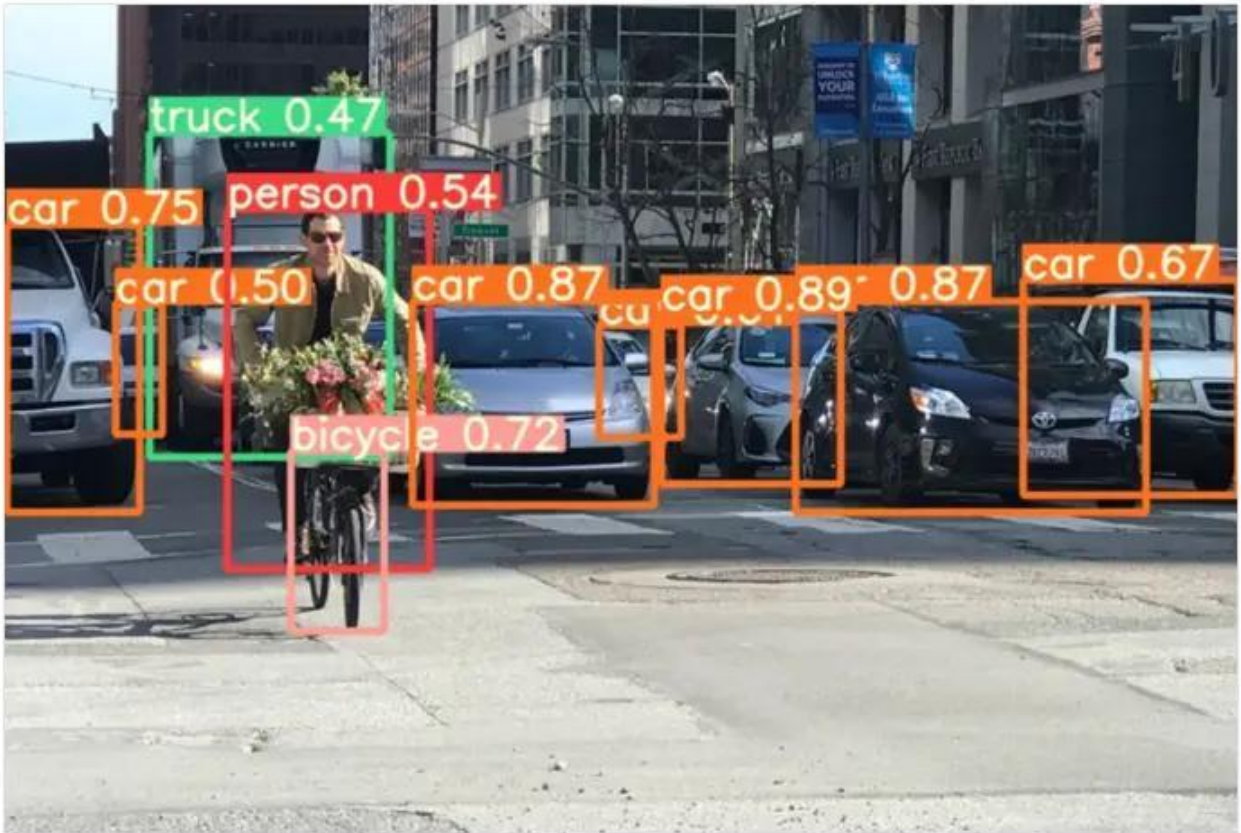
- Higher **precision**: 85.71% vs. 80%
- Higher **recall**: 81.82% vs. 72.73%
- More true positives: 90 vs. 80
- Fewer false positives: 15 vs. 20
- Fewer false negatives: 20 vs. 30

Final Selection: YOLO

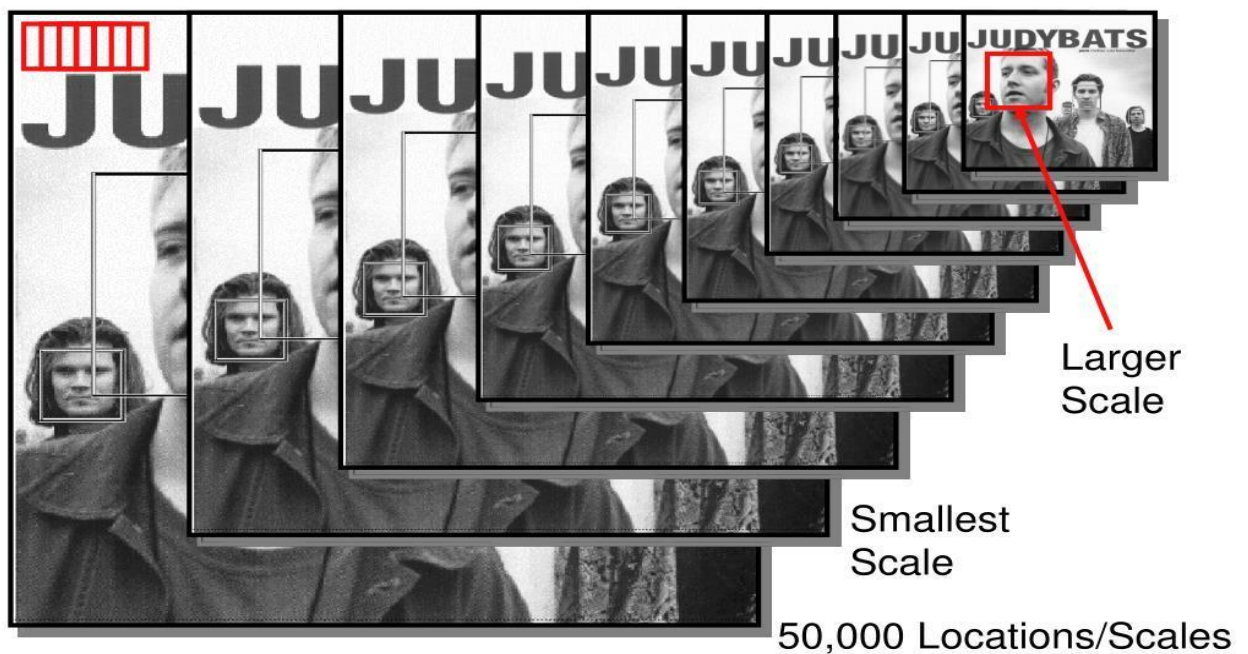
YOLO provides **better overall object detection performance** because it has both higher precision and higher recall.

Conclusion

YOLO is the better model because it detects more actual objects while producing fewer incorrect detections and fewer missed objects.



Viola/Jones: handling scale



2. Motion Estimation Error Analysis

Given Data

The displacement errors are:

- **Method A:** 2, 3, 2 pixels
- **Method B:** 3, 5, 4 pixels

(a) Calculate Average Displacement Error

Method A

Method B

Results

Method	Errors (pixels)	Average Error
Method A	2, 3, 2	2.33 pixels
Method B	3, 5, 4	4.00 pixels

(b) Comparison

- Method A has an average error of **2.33 pixels**.
- Method B has an average error of **4.00 pixels**.
- A smaller displacement error indicates more accurate motion estimation.

Final Selection: Method A

Method A provides **more accurate motion estimation** because its average error is lower.

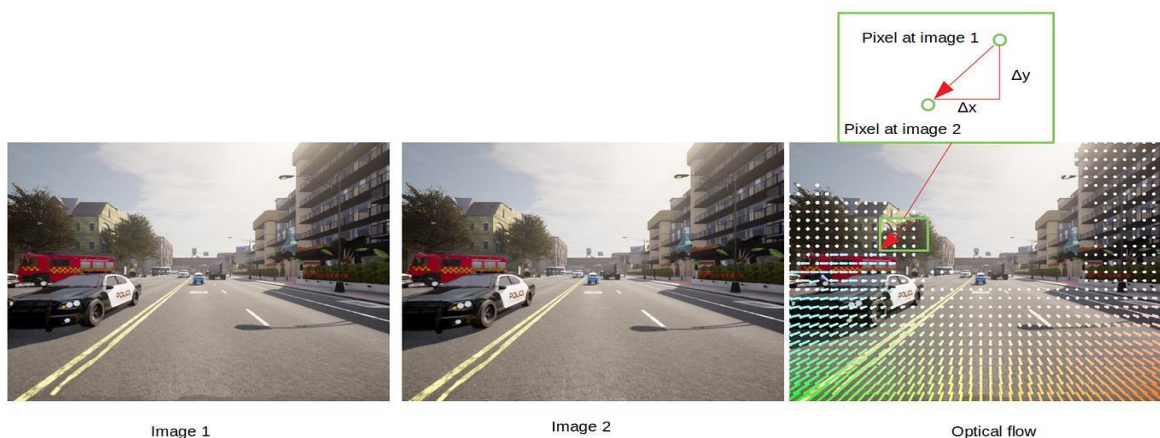
(c) Justification

Therefore, Method A has approximately **1.67 pixels less average error** than Method B.

Method A also has more consistent errors (**2, 3, 2**) compared with Method B (**3, 5, 4**).

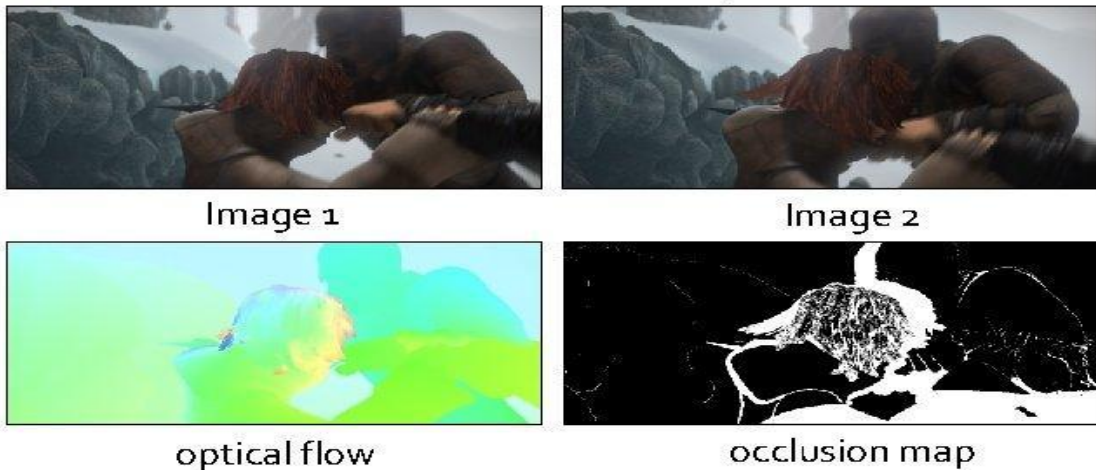
Conclusion

Method A is the better motion estimation technique because it produces lower and more consistent displacement errors.



Optical flow

□ Motion field between two image frames



3. Depth Estimation Comparison

Given Data

Object	Actual Depth	Stereo Vision	SfM
1	2.0 m	2.0 m	2.5 m
2	3.0 m	3.0 m	3.8 m
3	4.0 m	4.0 m	5.0 m

The formula for absolute error is:

(a) Absolute Error for Each Object

Error Comparison

Object	Actual	Stereo Error	SfM Error
1	2.0 m	0.0 m	0.5 m
2	3.0 m	0.0 m	0.8 m
3	4.0 m	0.0 m	1.0 m

(b) Average Absolute Error

Stereo Vision

SfM

Results

Method	Average Absolute Error
Stereo Vision	0.00 m
SfM	0.77 m

(c) Comparison and Final Selection

Stereo Vision provides more accurate depth estimation because:

- Its error is **0 m for all three objects**.
- Its average absolute error is **0 m**.
- Its estimated depths exactly match the actual depths in this dataset.

SfM has an average error of approximately **0.77 m**, and its estimated depth is higher than the actual depth for all three objects.

Final Selection: Stereo Vision

Conclusion

Stereo Vision is more accurate than SfM for the given data because it has zero absolute error, while SfM has an average absolute error of 0.77 m.

